

● Original Contribution

MOLECULAR ACOUSTIC ANGIOGRAPHY: A NEW TECHNIQUE FOR HIGH-RESOLUTION SUPERHARMONIC ULTRASOUND MOLECULAR IMAGING

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Abstract—Ultrasound molecular imaging utilizes targeted microbubbles to bind to vascular targets such as integrins, selectins and other extracellular binding domains. After binding, these microbubbles are typically imaged using low pressures and multi-pulse imaging sequences. In this article, we present an alternative approach for molecular imaging using ultrasound that relies on superharmonic signals produced by microbubble contrast agents. Bound bubbles were insonified near resonance using a low frequency (4 MHz) element and superharmonic echoes were received at high frequencies (25–30 MHz). Although this approach was observed to produce declining image intensity during repeated imaging in both *in vitro* and *in vivo* experiments because of bubble destruction, the feasibility of superharmonic molecular imaging was demonstrated for transmit pressures, which are sufficiently high to induce shell disruption in bound microbubbles. This approach was validated using microbubbles targeted to the $\alpha_v\beta_3$ integrin in a rat fibrosarcoma model ($n = 5$) and combined with superharmonic images of free microbubbles to produce high-contrast, high-resolution 3-D volumes of both microvascular anatomy and molecular targeting. Image intensity over repeated scans and the effect of microbubble diameter were also assessed *in vivo*, indicating that larger microbubbles yield increased persistence in image intensity. Using ultrasound-based acoustic angiography images rather than conventional B-mode ultrasound to provide the underlying anatomic information facilitates anatomic localization of molecular markers. Quantitative analysis of relationships between microvasculature and targeting information indicated that most targeting occurred within 50 μm of a resolvable vessel ($>100 \mu\text{m}$ diameter). The combined information provided by these scans may present new opportunities for analyzing relationships between microvascular anatomy and vascular targets, subject only to limitations of the current mechanically scanned system and microbubble persistence to repeated imaging at moderate mechanical indices. (E-mail: padayton@email.unc.edu) © 2016 World Federation for Ultrasound in Medicine & Biology.

Key Words: Microbubble, Targeted imaging, Angiogenesis, Microvasculature, Biomarker.

INTRODUCTION

In contrast to anatomical imaging methods, molecular imaging reveals functional information about tissue pathophysiology based on the accumulation of a molecular tracer. However, the value of a molecular imaging technique is often greatly enhanced by combining it with anatomic imaging in order to visualize the distribution of the marker. Clinically, this approach is used in systems

combining computed tomography (CT) with positron emission tomography (PET) or single photon emission computed tomography (SPECT). These combined systems have had higher accuracy in localizing abnormalities, often resulting in changes in treatment plans in cancers of the lung, thyroid, breast and prostate (Cerfolio et al. 2004; Garami et al. 2012; Lerman et al. 2007; Roach et al. 2006; Soyka et al. 2012; Tharp et al. 2004). In ultrasound molecular imaging, images of targeted microbubble contrast agent are typically overlaid on anatomic B-mode images to allow visualization of the spatial distribution of markers. Because of the intravascular nature of microbubbles,

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they are particularly well suited to the study of endothelial markers of disease. For this reason, many recent studies in ultrasound molecular imaging have investigated angiogenesis and related processes of vascular remodeling and inflammation (Anderson et al. 2011; Hyvelin et al. 2014; Inaba and Lindner 2012; Kaufmann and Lindner 2007; Voigt 2009; Wang et al. 2014).

Although ultrasound possesses advantages over other molecular imaging modalities because of its relatively low cost, portability and lack of ionizing radiation, it also has limited penetration depth at the high frequencies required to achieve sub-millimeter resolution. In addition, ultrasound imaging of anatomy is quite different in nature from CT or magnetic resonance imaging images that provide anatomic information for PET or SPECT, as contrast in B-mode ultrasound images is the result of differences in acoustic backscattering from tissue and depends on the size, distribution and acoustic impedance of the scatterers. Ultrasound also has a limited field of view compared to whole-body imaging modalities. The development of microbubble contrast agents, which provide substantially higher levels of acoustic scattering relative to erythrocytes, has greatly enhanced the ability to use ultrasound to image both vascular anatomy and blood-flow dynamics. Contrast-enhanced ultrasound imaging with high-frequency transducers has allowed imaging of small vessels with resolutions on the order of a few hundred microns, providing potential for assessing angiogenesis and vascular remodeling with both targeted and non-targeted microbubbles (Ellegala et al. 2003; Leong-Poi et al. 2005; Liu et al. 2008; Pysz et al. 2011; Wang et al. 2015a; Willmann et al. 2008, 2010). High-resolution molecular imaging is particularly valuable for pre-clinical imaging in small animal models and studies in which small anatomic locations are of interest, although high-frequency contrast-specific imaging is technically challenging to implement (Denbeigh et al. 2014; Foster et al. 2009, 2011; Liu et al. 2015; Rychak et al. 2007; Yan et al. 2012).

An alternative approach to high-frequency contrast-enhanced ultrasound involves the use of dual-frequency transducers, with a low-frequency transmit pulse used to excite microbubbles near their resonance and a high-frequency element receiving only higher harmonic echoes produced by microbubbles (Gessner et al. 2010). Detection of these “superharmonic” signals produced by microbubbles with multi-frequency transducers was first reported by Bouakaz et al. (2003) and by Kruse and Ferrara (2005), although these early investigators did not use this technology to develop the 3-D microvessel images attained more recently by Gessner et al. (2013). The “transmit low/receive high” dual-frequency approach has previously been utilized in molecular imag-

ing by Ferrara and colleagues (Hu et al. 2010, 2013). When transmitting at 2 MHz and receiving at 15 MHz, Hu et al. (2013) found approximately a two-fold improvement in spatial resolution *in vitro* relative to multi-pulse techniques employed by commercial scanners. The use of a low transmit frequency improves the penetration depth relative to contrast imaging at a single high frequency. For example, at a depth of 1.5 cm and an attenuation of 0.2 dB/cm/MHz, two-way attenuation in a 25 MHz scan is 15 dB, but is only 8.7 dB with a 4/25 MHz dual-frequency scan. Transmitting at a lower frequency also more closely corresponds to the resonance frequencies of most commercially available microbubble contrast agents (Doinikov et al. 2009; Faez et al. 2011; Helfield and Goertz 2013), improving image contrast, which is vital in molecular imaging when relatively few microbubbles are typically present.

Using this “transmit low/receive high” approach with prototype dual-frequency, mechanically steered transducers connected to a commercial ultrasound system, we have recently demonstrated the ability to form high-resolution images of vasculature with resolutions of approximately 150 μm and with almost no tissue interference. We call this technique “acoustic angiography” because of the images’ similarity to CT or MR angiography. Although commercial ultrasound scanners perform contrast-specific imaging within a single transducer bandwidth, our approach forms images using only received signals at frequencies at least three times higher than the transmitted frequency. Our group recently found that using a receiving center frequency at least three times higher than the transmit center frequency produces the highest contrast-to-tissue ratios, as tissue amplitudes decrease more rapidly than microbubbles’ amplitudes with increasing frequency (Lindsey et al. 2014). In addition, although using higher harmonics to form images (*i.e.*, 5th rather than 3rd) may further reduce sensitivity, it provides the increased resolution necessary for resolving vasculature <300 μm in diameter. Using these 3-D imaging volumes and quantitative metrics of vascular signatures, we have further demonstrated the ability to distinguish between healthy and cancerous tissues (Gessner et al. 2012a; Shelton et al. 2015). High specificity to contrast resulting from transmitting at low frequencies and receiving at much higher frequencies allows suppression of tissue artifacts while preserving frame rates without requiring a separate control image to be acquired and subtracted. This simplifies the imaging protocol and eliminates artifacts due to physiologic motion or mis-registration of the control and contrast images.

Several studies have found that the sensitivity of ultrasound molecular imaging may also be improved by adjusting the size distribution of the microbubble

population (Sirsi *et al.* 2010; Streeter *et al.* 2010; Talu *et al.* 2007). In dual-frequency superharmonic imaging, we have recently observed that using a population with larger microbubbles results in superharmonic signals with increased amplitude (Lindsey *et al.* 2014). This effect arises because larger microbubbles are capable of producing superharmonic signals over a greater number of acoustic pulses (Lindsey *et al.* 2015).

Despite the promising resolution of dual-frequency superharmonic imaging, our own recent studies indicate that dual-frequency superharmonic imaging is at least partially destructive to microbubbles (Lindsey *et al.* 2015), which makes implementation of superharmonic molecular imaging a significant challenge. The dual-frequency transducer design is critical for separation of tissue and microbubble echoes, as a wide separation between transmit and receive bandwidths ensures that only microbubble echoes are received at the higher frequencies. However, the focal spot is larger for the low-frequency element than the high frequency element. The result is that as both transducers are swept in unison to form a 2-D image, a stationary microbubble is subjected to multiple off-axis, low-frequency transmit pulses before the center of the dual low- and high- frequency focus is aligned to the microbubble. Although many commercially available systems utilize techniques such as synthetic aperture processing and broad beam insonation to ensure uniform acoustic intensity over wide regions of interest for microbubble imaging (Brock-Fisher *et al.* 1996), the transducers utilized by these systems do not have the necessary bandwidth to perform the superharmonic contrast imaging enabled by the described prototype transducer.

Although the mechanical indices used in superharmonic imaging are below the maximum of 0.8 for which safety has been evaluated for Definity (Lantheus Medical Imaging, 2013), our previous studies indicate that microbubble shell disruption is required to produce superharmonic signals (approximately 300 kPa or greater at 4 MHz; Lindsey *et al.* 2015), which may present a challenge for molecular imaging if bound microbubbles might be destroyed before the superharmonic signal they produce can be detected. Fortunately, there is evidence that the same microbubbles can produce superharmonic echoes in response to multiple insonations (Kruse and Ferrara 2005; Lindsey *et al.* 2015), suggesting that it may be possible to balance production of superharmonic signals and microbubble destruction in such a way as to enable molecular imaging.

In this article, we describe the development of an approach for high-resolution molecular imaging based on microbubble superharmonic echoes. This technique uses spectrally separated transmit and receive bandwidths to form images of bound microbubbles rather than the

low mechanical index, multi-pulse sequences that are commonly used (Phillips 2001). Intensity and persistence of the superharmonic signals in these images were evaluated through *in vitro* experiments in micro-cellulose tubes and *in vivo* imaging studies in a subcutaneous tumor model. When combined with a sequentially acquired acoustic angiography volume, the developed approach provides simultaneous 3-D visualization of both microvascular structure and molecular markers of angiogenesis with unprecedented resolution. Whereas the B-mode ultrasound images that typically provide anatomic information in ultrasound molecular imaging are difficult to display in 3-D and require an experienced user to understand and interpret differences between tissue types, the vascular anatomy provided by acoustic angiography can be easily displayed in 3-D and may be readily interpreted by users with limited previous ultrasound experience. Finally, microvascular information in the acoustic angiography volume can be analyzed using metrics such as vascular density (Dunleavy *et al.* 2014) or tortuosity to quantitatively assess vascular remodeling (Gessner *et al.* 2012a; Shelton *et al.* 2015). When combined with molecular targeting information, this may provide new information regarding the spatial distribution of vascular biomarkers.

MATERIALS AND METHODS

Acoustic angiography imaging experiments were performed with modified VisualSonics RMV probes and a Vevo 770 ultrasound scanner (VisualSonics, Toronto, Canada). In this work, modified RMV710 B (25-MHz center frequency) and RMV707 (30-MHz center frequency) transducers were used, each with an additional 4-MHz annular element that is confocal to the high-frequency element (Gessner *et al.* 2010). The low frequency element is used only for transmitting, whereas the high frequency element only receives higher harmonics produced by microbubbles. The two single-element transducers are mechanically steered to acquire a 2-D image and can be translated using a motion stage to combine multiple 2-D images into volumetric data.

Microbubble preparation for in vitro studies

Lipid solutions (Streeter *et al.* 2010) were formulated with a 9:1 molar ratio of 1,2-distearoyl-*sn*-glycero-3-phosphocholine (DSPC) and 1,2-distearoyl-*sn*-glycero-3-phosphoethanolamine-N-methoxy (polyethylene-glycol)-2000 (DSPE-PEG2000; Avanti Polar Lipids, Alabaster, AL, USA) in a solution containing propylene glycol 15% (v/v), glycerol 5% (v/v) and phosphate-buffered saline (PBS) 80% (v/v). Next, 1.5-mL aliquots of this lipid solution were placed in sealed 3-mL glass vials and the air headspace was exchanged with decafluorobutane gas

purchased from Fluoromed (Round Rock, TX, USA). Microbubbles with decafluorobutane gas cores and phospholipid shells formed spontaneously when agitated in a Vialmix device (Lantheus Medical Imaging, N. Billerica, MA, USA). The concentration and diameter of microbubbles in this emulsion was measured *via* (single particle) optical techniques (Accusizer 780, Particle Sizing Systems, Santa Barbara, CA, USA; [Satinover et al. 2014](#)). For targeted bubbles, a biotinylated PEG was used (DSPE-PEG2000-Biotin; [Zhao et al. 2004](#)) to replace 50% of the DSPE-PEG2000. Microbubble emulsions were washed with four centrifugations and produced a stable population of microbubbles with a diameter of approximately 1.4 μm ([Feshitan et al. 2009](#)).

In vitro experiments

Transducer calibration and spatial mapping of the low-frequency transmit beam was performed with a calibrated needle hydrophone (HNA 0400, Onda Corp., Sunnyvale, CA, USA) on a computerized motion stage (Newport XPS, Irvine, CA, USA) and digitized using a digital acquisition board (Signatec PDA14, Corona, CA, USA). The hydrophone was positioned at the focus of the transducer and translated laterally to measure a 4-mm extent of the beam profile at the focal depth (15 mm for RMV 710 B). In acoustic angiography, the resolution is primarily dictated by the high-frequency receiving transducer, and we have recently characterized axial resolution for microbubble superharmonic imaging across a wide range of frequencies ([Lindsey et al. 2014](#)). Relative to the study of [Hu et al. \(2013\)](#), our own data indicate that axial resolution might be expected to decrease from $\sim 250\ \mu\text{m}$ at frequencies of 2/15 MHz to $\sim 170\ \mu\text{m}$ at 4/25 MHz, providing slightly improved resolution, though the contrast-to-tissue ratio is expected to decrease with increasing receiving frequency. Therefore, line spacing in a 2-D image is determined by the resolution of the high-frequency element. Manufacturer-provided information on scan geometry was used to compute line spacing at the focus (43 μm) to determine the overlap of the low-frequency transmit pulse in adjacent lines.

In vitro microbubble targeting was achieved using biotinylated microbubbles and an avidin-coated cellulose tube. First, the 200- μm (inner diameter) cellulose tube (Spectrum Labs, Rancho Dominguez, CA, USA) was infused with 100% ethanol, which was then cleared by filling the tube with air. Next, the tube was filled with a 5-mg/mL solution of avidin (Sigma-Aldrich, St. Louis, MO, USA) in PBS and allowed to sit at room temperature for 30 min before being flushed with PBS and positioned in a water bath at the focus of the dual-frequency transducer. An infusion of 5×10^4 biotin-coated microbubbles in 300 μL of PBS was pumped through the tube at a rate

of 30 $\mu\text{L}/\text{min}$ (Harvard Apparatus PHD2000, Holliston, MA, USA), followed by a 10-min flush of pure PBS to remove any unbound microbubbles. Two-dimensional images of the bound microbubbles were acquired for 100 frames at a frame rate of 1 Hz at peak negative pressures of 200, 350 and 500 kPa. Then, image clips were exported and analyzed in ImageJ (National Institutes of Health, Bethesda, MD, USA). An image of the tube before contrast infusion was subtracted from each set of acquired images to eliminate any non-microbubble artifacts, and then a region of interest was defined around the tube to compute image intensity. Experiments were repeated in triplicate and results were averaged together for each transmit pressure tested. A new tube was prepared with a new microbubble injection for each trial.

Microbubble preparation for in vivo studies

Microbubbles targeted to the $\alpha_v\beta_3$ integrin were created with an 18:1:1 molar ratio of DSPC, DSPE-PEG2000 and DSPE-PEG2000-maleimide cross-linked to the peptide Cyclo-Arg-Gly-Asp-D-Tyr-Cys (cyclic RGD, Peptides International, Louisville, KY, USA) in PBS (Fisher Scientific, Pittsburgh, PA, USA). The cyclic RGD peptide targets vessels expressing $\alpha_v\beta_3$ integrin, which is characteristic of angiogenic tumors ([Anderson et al. 2011](#); [Dayton et al. 2004](#); [Ellegala et al. 2003](#); [Hu et al. 2013](#); [Leong-Poi et al. 2003](#); [Streeter et al. 2010, 2011](#)).

Previous investigations of the relationship between microbubble diameter and molecular imaging ([Streeter et al. 2010](#)) and contrast-to-tissue ratio in acoustic angiography imaging ([Lindsey et al. 2014](#)) both demonstrated improved performance for bubbles having diameters in the range of 3–4 μm rather than 1–2 μm . Additionally, larger bubbles are likely to produce superharmonic signals over a greater number of pulses than smaller bubbles ([Lindsey et al. 2015](#)). To examine the effect of microbubble diameter on superharmonic imaging *in vivo*, two populations of microbubbles were prepared: a polydisperse population as described in our *in vitro* studies with a mode diameter of approximately 1.4 μm , and larger-diameter microbubbles enriched using differential centrifugal separation ([Feshitan et al. 2009](#); [Streeter et al. 2010](#)) with a mode diameter of approximately 3.8 μm .

Animal care and imaging

The Fisher 344 fibrosarcoma model was prepared as previously described ([Yuan et al. 2006](#)). *In vivo* imaging occurred when tumors reached approximately 1 cm in diameter. Rats were anesthetized with vaporized isoflurane and oxygen, and a 24-gauge catheter was inserted into the tail vein. Fur was removed from the imaging region by shaving and chemical depilation, and ultrasound gel was applied to the skin. Body temperature

was maintained throughout using a heated imaging stage, and rats were secured to the imaging platform with surgical tape to minimize motion during the study. Acquisition times were less than 1 min for the molecular volume and between 1 and 2 min for the microvascular volume. All animal experiments were performed in accordance with The University of North Carolina at Chapel Hill Institutional Animal Care and Use Committee.

Using the same VisualSonics 770 system and prototype transducer, tumor location was first identified in high-frequency B-mode imaging. Next, a dual-frequency image was acquired before contrast agent administration for identification of any tissue artifacts. A bolus of 5×10^7 cyclic RGD microbubbles was injected through the tail-vein catheter. After a 12-min wait, circulating microbubbles had cleared, and 3-D images of targeted microbubbles were acquired in dual-frequency mode with a step size of $250 \mu\text{m}$ between 2-D images to avoid destroying bubbles in the adjacent imaging plane. The 12-min wait for targeting is similar to that used in previous ultrasound molecular imaging studies (Inaba and Lindner 2012; Kaufmann *et al.* 2007; Wang *et al.* 2015b) and was determined based on prior scanning of two animals in which the degree of free bubbles in circulation was assessed every 4 min post-injection for 28 min. To test the expected improvement due to increased microbubble diameter, molecular imaging scans were acquired consecutively with size sorted (larger) and polydisperse (smaller) microbubbles in the same animal, with any remaining bound microbubbles destroyed between bolus injections by performing a scan at a peak negative pressure of 1400 kPa. The order in which the pressures were tested was randomized in order to mitigate any potential effects of repeated boluses, which are expected to be negligible (Streeter and Dayton 2013). To examine microbubble destruction over repeated imaging, four 3-D scans were acquired at each set of parameters (microbubble diameter, pressure). After the acquisition of molecular images, conventional dual-frequency acoustic angiography images were acquired using a constant infusion of 1.5×10^8 untargeted microbubbles/min, a peak negative pressure of 1.2 MPa and a step size of $100 \mu\text{m}$ between 2-D images.

Image processing and analysis

Volumetric image data of targeted microbubbles were imported into ImageJ and background noise was reduced by subtracting the background using the “rolling ball” algorithm with a radius of 30 (Sternberg 1983). Tissue signal was reduced by subtracting a volume acquired before contrast agent injection. In MATLAB (The MathWorks Inc., Natick, MA, USA), thresholding was applied to remove any remaining noise. The image intensities in volumetric tumor regions were calculated for each scan,

and the average intensities of successive scans were normalized to the first scan to compare the relative loss of signal due to microbubble destruction. Acoustic angiography and molecular imaging volumes for the same animals were combined into a single volume in ImageJ by inserting blank frames into the molecular imaging volumes so that the slice spacing matched that of the acoustic angiography volumes, then merging the two volumes using different colors, with acoustic angiography data displayed in white and targeting information displayed in green.

For five animals, individual vessels were segmented from acoustic angiography image volumes using center-line extraction *via* height ridge traversal with manually defined seed points (Aylward and Bullitt 2002). Coordinates of segmented vessels, as well as coordinates and voxel intensities of the targeting sites, from the molecular imaging data were read into MATLAB. Vessel coordinates were down-sampled to match the image spacing of the molecular imaging data. In order to analyze spatial relationships between resolvable vessels and targeted microbubble signal, distances between the targeting site and the five closest vessels were computed for each targeting site. Any registration artifacts due to differences in respiratory motion between sequential scans were assumed to be negligible. Previous studies indicate that the mean root-mean-square error between two consecutive frames due to physiologic motion is $7.2 \pm 3.3 \mu\text{m}$. Vessel diameters for these five closest vessels were also recorded for each targeting site.

RESULTS

In vitro

Transmit pressure profiles were created at peak negative pressures of 200, 350 and 500 kPa at 4 MHz (Fig. 1a) to visualize pressure levels incident on neighboring microbubbles from a single low-frequency transmit event. With a mechanically steered transducer, a single microbubble in the field is subject to increasing pressures from the low-frequency element as the transducer arm moves toward the microbubble. Thus, by the time both the peak transmit pressure and the focus of the high-frequency receive element are aligned to the microbubble for acquisition of the imaging line, the microbubble has already experienced several pulses at increasing pressures. Figure 1b contains a diagram describing the motion of the transducer across 10 imaging lines occupying $388 \mu\text{m}$ of the image in the lateral direction (only three transducer positions are shown instead of 10 for clarity). A single microbubble positioned at the location of line 10 receives gradually increasing pressures with each transmit pulse as the transducer moves from line 1 to line 10. Figure 2 depicts this overlapping spacing

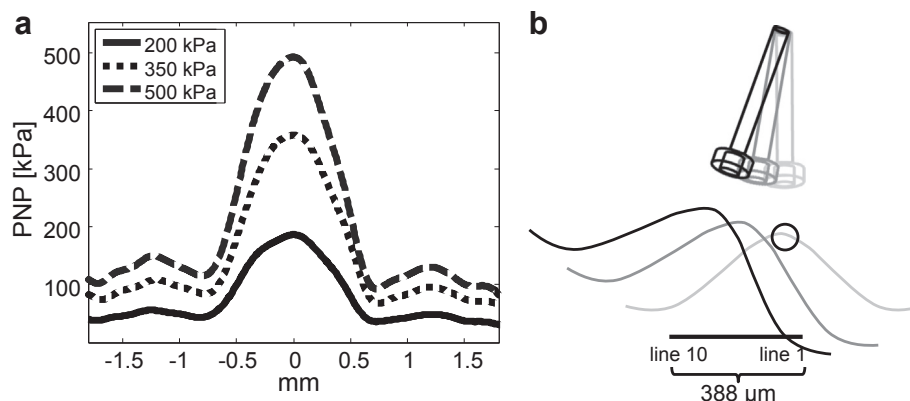


Fig. 1. (a) Hydrophone-measured pressure maps as a transmit beam having a peak negative pressure of 200 kPa (solid line), 350 kPa (dotted line) or 500 kPa (dashed line) is swept from left to right. (b) Motion of transducer mechanical arm for the scenario in (a). Illustration of transmitted pressure experienced by a single microbubble as the mechanically steered transducer arm moves from some distance away from the bubble (black) to on-axis steering (gray). Given a line separation of $43.1\ \mu\text{m}$, 10 adjacent lines occupy a distance of $387.9\ \mu\text{m}$, over which a single bubble is exposed to multiple pulses of gradually increasing pressure until the on-axis bubble experiences the nominal peak negative pressure.

of the 10 low-frequency transmit pulses for the 350 kPa case. Vertical lines delineate the transmit line spacing of $43.1\ \mu\text{m}$ and horizontal lines show the magnitude of negative pressure experienced by a bubble at location zero (in focus at line 10) for preceding transmit events.

Bound microbubbles in cellulose tubes were observed over 100 frames to assess frame-to-frame decrease in signal intensity because of microbubble destruction by the transmit pulses (Fig. 3). As expected, initial intensity of the targeted microbubble signal is directly related to the peak negative pressure of the transmit pulse. All three pressures (200, 350 and 500 kPa, peak negative pressure) experienced some loss of signal from the initial transmit pulse, likely due to the preferential destruction of small microbubbles (Lindsey et al. 2015). However, observations at 200 and 350 kPa indicate relatively slow decline in the microbubble signal, with a greater decrease in intensity observed for the 500 kPa

test case. At the lowest pressure (200 kPa), the bound microbubble signal remained relatively stable across the 100 transmit pulses, indicating minimal destruction. However, the intensities of the superharmonic echoes received by the high-frequency element were low for the 200 kPa transmit pressure case. At the highest pressure (500 kPa), average intensity decreased quickly in the first few frames and then decayed slowly over approximately frames 10–100. With the moderate pressure (350 kPa), the intensity decreased gradually over all 100 frames and exhibited greater microbubble signal than the observations at 200 and 500 kPa by frame 3. Overall, *in vitro* results indicated that a balance between microbubble destruction and generation of broadband superharmonic signals might be achievable for bound microbubbles for *in vivo* molecular imaging.

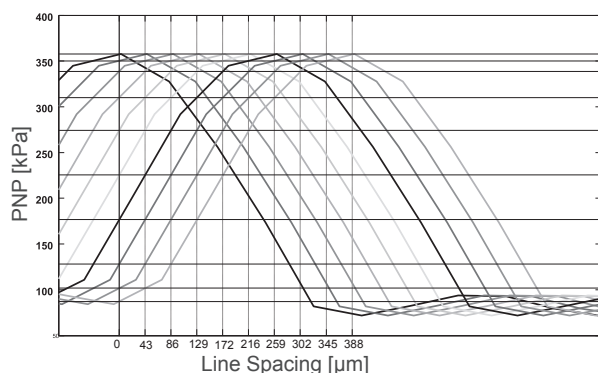


Fig. 2. Peak negative pressures (horizontal lines) in kPa experienced by a bubble at position 0, due to 10 adjacent scan lines with a line spacing of $43.1\ \mu\text{m}$ (illustrated by the vertical lines).

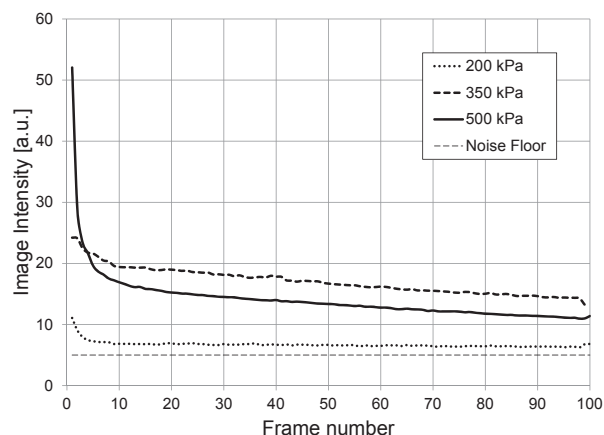


Fig. 3. Image intensity vs. frame number for targeted biotin microbubbles adhered to an avidin-coated tube. Each line indicates the mean result of three experiments.

In vivo

Imaging of microbubbles targeted to the $\alpha_v\beta_3$ integrin found little to no detectable signal *in vivo* at the low peak negative pressures examined *in vitro*. Therefore, the pressures tested were increased to non-detracted values of 700, 900 and 1200 kPa. Several reasons account for the need for greatly increased pressures *in vivo*, including attenuation, small vessel confinement effects and blood viscosity. For example, for an imaging depth of 15 mm, an attenuation of 0.2 dB/cm/MHz in overlying tissues represents a loss of 1.2 dB at 4 MHz. Superharmonic signals from the relatively few bubbles under the high-frequency focus are also difficult to detect at the transducer, as the same attenuation of 0.2 dB/cm/MHz results in attenuations of 9 dB at 30 MHz. Additionally, Caskey *et al.* (2007) have reported that microbubble oscillation is constrained when bubbles are confined to smaller vessels *in vivo*, measuring a relative expansion of a factor of 4–11.5 in 200- μ m inner diameter tubes *in vitro* compared with a mean relative expansion of 1.5 in *ex vivo* vessels less than 30 μ m in diameter. This may indicate higher pressures are required to produce superharmonic signals in small vessels. Other authors have investigated the effect of blood viscosity on microbubble oscillation, finding that microbubble resonance is severely altered in the presence of viscous damping (Khismatullin 2004; Sassaroli and Hynynen 2004).

With these higher pressures, it was possible to create *in vivo* images of bound microbubbles that could be overlaid on acoustic angiography image volumes as a reference for vascular structure. Note that these molecular imaging volumes could also be overlaid on B-mode images, as is customary in ultrasound molecular imaging.

Figure 4 shows fused maximum intensity projections of molecular and acoustic angiography images in the lateral view (4A) and dorsal-ventral view (4B) of a tumor approximately 1 cm in diameter, using a polydisperse population of microbubbles (1–2 μ m diameter) and a peak negative pressure of 1200 kPa. The relative decrease in image intensity over four successive scans in the same five animals for both sorted or polydisperse microbubble populations is shown in Figure 5. During the first scan, the larger sorted microbubbles were clearly brighter than the first scan of the polydisperse population, but the difference between the signal from each population decreased through scan 4, when neither microbubble population yielded a significant detectable signal at a transmit peak negative pressure of 1200 kPa.

In comparing image intensity for the size-sorted microbubbles at pressures of 700, 900 and 1200 kPa in the same animal, the higher pressures were found to produce the greatest image intensity from bound microbubbles. Figure 6 shows example image overlays of acoustic angiography and molecular imaging at the three pressures tested in a single representative animal. At higher pressures, both the brightness of individual targeting sites and the number of detectable targeting sites were visibly greater. The relative image intensities of the three different peak negative insonation pressures were computed across four scans until very little signal was detectable at any pressure, and these data are shown in Figure 7 for all five animals. The plot in Figure 7 shows that the mean image intensity in targeted microbubble images at 700 kPa is fairly constant over four scans, although the mean image intensity value is low. At higher pressures, the mean initial intensity of scan 1 improved,

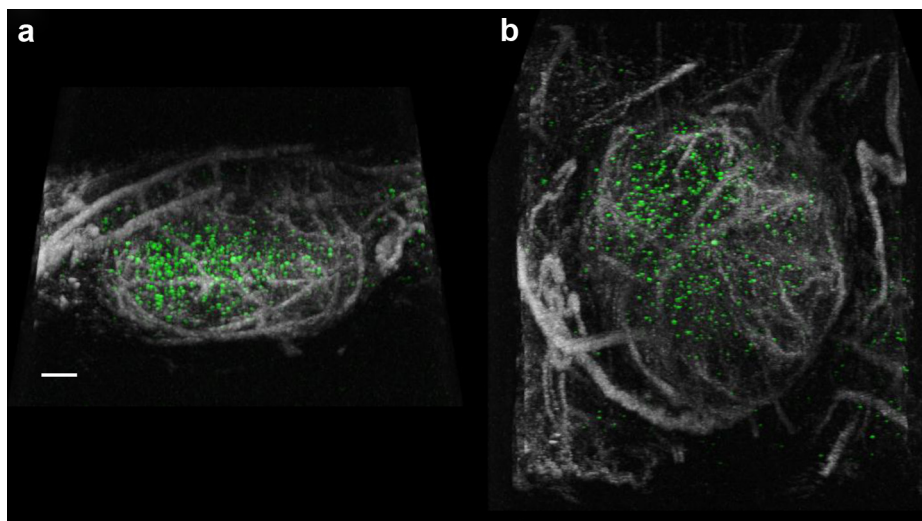


Fig. 4. (a) Fused scans of vascular anatomy acquired using untargeted (gray-scale) and cyclic RGD microbubbles (green) in sequential scans of a subcutaneous fibrosarcoma in the same animal. Images are shown here in a lateral view. (b) The same data in (a) are shown in a dorsal-ventral view. Both scale bars represent 2 mm.

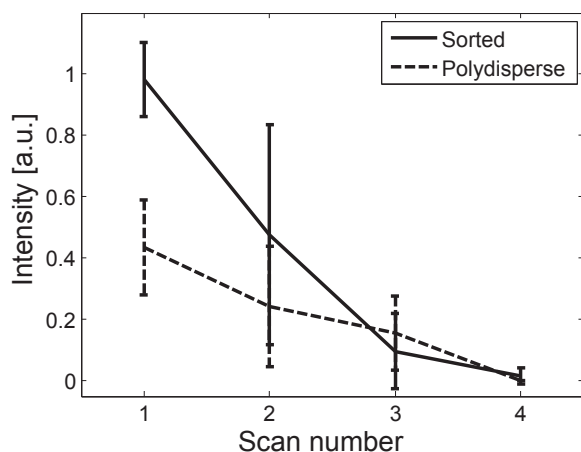


Fig. 5. Image intensity over four consecutive scans with cyclic RGD bubbles in five animals at non-derated peak negative transmit pressures of 1200 kPa for both sorted and polydisperse cyclic RGD microbubble populations. Each scan was normalized to its initial intensity.

but there was a greater decline in image intensity with each successive scan. The greatest targeting signal intensity resulted from the first scan at 1200 kPa, the highest peak negative pressure tested.

Segmentation of vessels visible in acoustic angiography images acquired with a constant infusion of microbubbles allowed comparison of the spatial distribution of targeted microbubbles and vascular structure. After segmentation, distances from each targeting site to the five nearest vessels and the diameters of those vessels were computed in MATLAB (see Fig. 8). The majority of targeting sites were within 50 μm of the centerline of a resolvable vessel, and very few targeting sites were located farther than 150 μm from a segmented vessel (Fig. 8a). Additionally, we observed that the majority of targeting occurred in small vessels. Figure 8b shows that $\alpha_v\beta_3$ targeting increased as vessels decreased in diameter, with a decline in the percentage of targeting sites located in or near vessels 100–200 μm in diameter, which is approximately the resolution limit of the acoustic angiography images. The relationship between vessel diameter and levels of targeted microbubble detection may be related to purely physical implications of smaller diameter vessels, or to patterns of $\alpha_v\beta_3$ integrin expression.

DISCUSSION

Bubble destruction and superharmonic molecular imaging

In this article, we have demonstrated the feasibility of forming high-resolution images of microbubbles targeted to the $\alpha_v\beta_3$ integrin using an imaging technique that forms images of only the superharmonic echoes of

microbubbles. Although this technique is known to be partially destructive, bound microbubbles persisted sufficiently long *in vivo* to allow for the formation of high-resolution superharmonic molecular images (Fig. 4). While image intensity decreased rapidly with each subsequent scan (Figs. 5 and 7), superharmonic echoes were detectable on the initial scan despite the fact that microbubbles had already been subjected to multiple pulses due to the large focal size of the low-frequency transducer element (Figs. 1 and 2). This partial persistence of superharmonic signals over a few pulses is consistent with the *in vitro* studies reported here (Fig. 3) and previously (Lindsey et al. 2015). We have also observed that superharmonic molecular imaging may be enhanced by using a microbubble population with a slightly larger diameter, consistent with previous observations that larger microbubbles improve sensitivity in conventional molecular imaging (Streeter et al. 2010; Talu et al. 2007) and in superharmonic imaging of free bubbles (Lindsey et al. 2014), and that larger free microbubbles frequently deflate over several pulses relative to smaller microbubbles (Lindsey et al. 2015). Although microbubble persistence is clearly a challenge for this technique, acquiring useful superharmonic molecular images is possible on at least the first acquisition and appears to be aided by the use of microbubbles having slightly larger diameters. In addition, the use of higher microbubble concentrations might produce increased targeting more similar in appearance to conventional ultrasound molecular imaging approaches, but the current concentration allows spatial comparison between molecular and microvascular imaging volumes.

Potential improvements

Although superharmonic molecular images with the highest image intensity were acquired with the highest pressure tested (1200 kPa), opportunities may remain for further optimization of transmit frequencies or pressures to either increase sensitivity of the initial scan or prolong the usefulness of the images over a greater number of scans. The effects of the *in vivo* environment on microbubble oscillation have not been precisely quantified, so the peak negative pressure required to induce microbubble deflation or destruction *in vivo* are unclear. Previous studies have reported that microbubbles persist longer as vessel size decreases (Caskey et al. 2007); however, greater pressures may be required to produce superharmonic echoes from these bubbles. This suggests that in a field of view filled with vessels of varying diameters, microbubble oscillation and thus received signal may depend on vessel diameter. As such, sensitivity to smaller vessels may not be limited merely by the presence of fewer microbubbles or slower re-perfusion rates in these vessels.

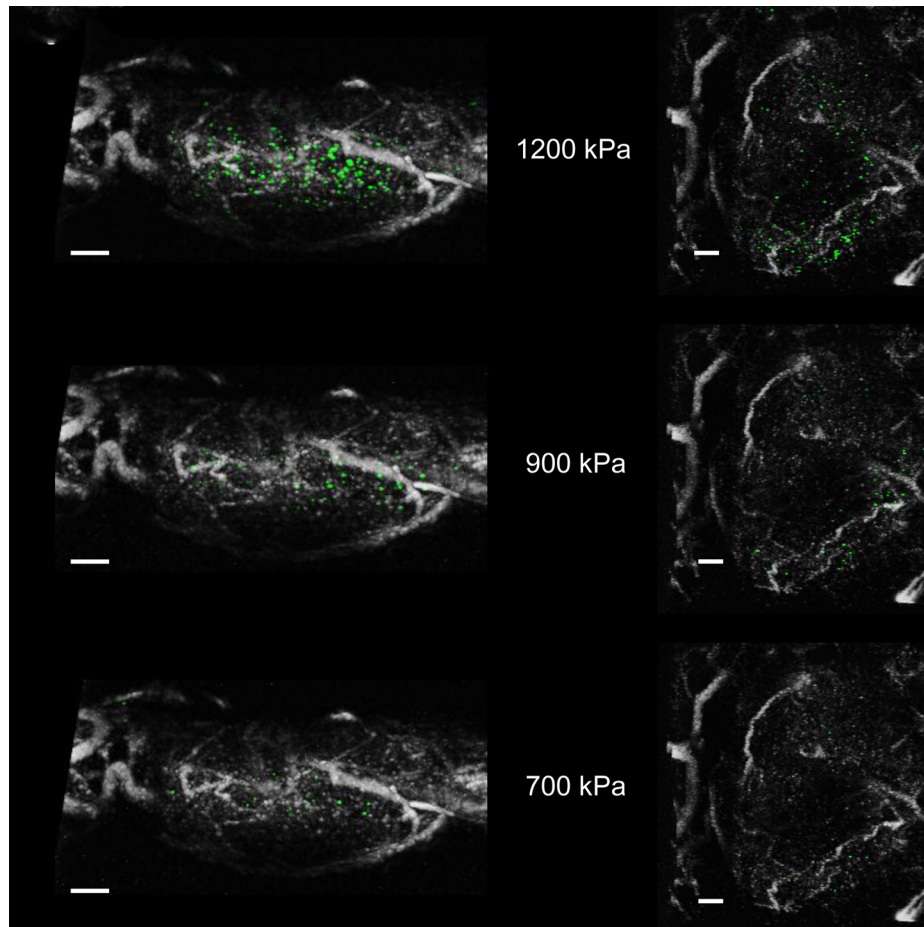


Fig. 6. The effect of varying peak transmit pressure on received molecular imaging signal (*green*) is illustrated in a single animal in both lateral (*left*) and dorsal-ventral views (*right*). The same acoustic angiography data (*gray-scale*) were used to provide vascular anatomy for all cases.

Because of the mechanical steering of the prototype transducer, each microbubble is subjected to many lower-pressure pulses before the receiving focus is aligned over the microbubble. Additionally, bound microbubbles are subject to secondary Bjerknes forces, which may cause detachment of bound microbubbles at low mechanical index (Garbin *et al.* 2011; Gessner *et al.* 2012b; Loughran *et al.* 2012). These forces may induce microbubble unbinding in this method due to lower pressure pulses acting on a microbubble before the dual focus is aligned with the microbubble, potentially reducing sensitivity if detached microbubbles flow out of plane before being detected. Both of these effects (destruction and detachment of microbubbles outside of the receiving focus) could be mitigated by using a transducer having transmit and receive foci that were more similar in size, as the lateral resolution of the high-frequency element is $140\ \mu\text{m}$ (710 B) whereas that of the low-frequency element is $\sim 1\ \text{mm}$. Alternatively, a dual-frequency array transducer would allow all echoes

to be received by the high-frequency array even with a large transmit focus (Bouakaz *et al.* 2002; Martin *et al.* 2014; Stephens *et al.* 2006).

Use of an array would also help mitigate any effects of motion artifacts in the spatial comparison between vascular and molecular imaging data by allowing increased acquisition rates and providing a more uniform acoustic field. However, because vessel centerlines are extracted using the described segmentation technique, single frame shift errors due to animal respiration are removed from microvasculature imaging data. This means that only the molecular imaging data is subject to the effects of motion artifacts. However, the fact that results indicate meaningful spatial relationships between vessel diameter and distance to targeting sites (Fig. 8) suggests that any motion effects in the molecular imaging data sets are sufficiently small as to permit this type of analysis with the current system.

The use of chirped excitation may be an alternative means for improving sensitivity to bound microbubbles,

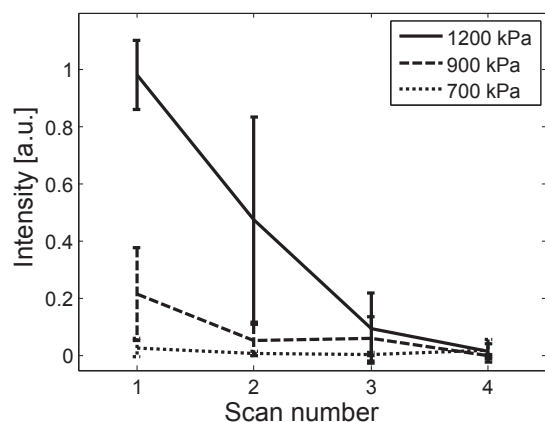


Fig. 7. Image intensity over four consecutive scans with cyclic RGD bubbles in five animals with varying peak negative transmit pressures of 700, 900 and 1200 kPa for size-sorted microbubble populations. Each scan was normalized to its initial intensity.

which has previously been demonstrated in contrast imaging (Borsboom et al. 2005; Leavens et al. 2007; Sun et al. 2006, 2007). Although this may enable molecular imaging of microbubbles of varying diameters at lower peak pressures, these pulses have not been assessed in superharmonic contrast imaging, in which the highest amplitude echoes are produced by microbubble destruction (Lindsey et al. 2015). This suggests that short pulses may be more effective for superharmonic imaging because they induce microbubble destruction over a smaller volume than longer pulses. Use of chirped pulses also depends on the ability of the transducer to transmit a pulse without distortion (Lindsey et al. 2015).

Value of combined molecular and vascular imaging

The combination of molecular imaging and acoustic angiography provides opportunities for investigating the spatial distribution of functional biomarkers and vascular structures in concert. Given that ultrasound molecular

imaging utilizes endothelial targets, combining ultrasound molecular imaging with angiography images mapping vascular structure provides the unique opportunity to analyze relationships between vascular features and molecular information. In this work, we have observed that cyclic RGD microbubbles targeted to the $\alpha_v\beta_3$ integrin were detected preferentially in smaller vessels, and that all targeting sites were located within 200 μm of a resolvable vessel, with the majority of targeting occurring within 50 μm of a segmented vessel. Molecular targeting may be increased in small vessels because of slower blood flow velocities or the increasing similarity between microbubble and vessel diameters, which may improve the probability of binding (Kaufmann et al. 2007; Takalkar et al. 2004). Alternatively, binding in larger vessels may prove transient or more challenging due to detachment caused by greater shear flow, with binding efficiency in shear flow also dependent upon ligand concentration on the microbubble surface (Klibanov et al. 2006; Patil et al. 2011). When microbubbles detach due to either shear flow or ultrasound, these microbubbles may be retained by smaller vessels for a longer time period than in larger vessels.

However, straightforward relationships between vessel diameter and flow rate do not hold in tumors because of the formation of abnormal vascular networks created by tumor angiogenesis (Jain 1988). The spatial distribution of integrin expression may also vary with vessel diameter, as previous investigations have reported that molecular targets of angiogenesis, such as vascular endothelial growth factor or the $\alpha_v\beta_3$ integrin used in this work, may be more prevalent in small vessels undergoing active remodeling (Brooks et al. 1994). Although the exact relationship between vessel anatomy and molecular targeting is unclear, the presented approach has demonstrated an initial ability to assess the sizes of vessels in which targeting occurs in pre-clinical studies.

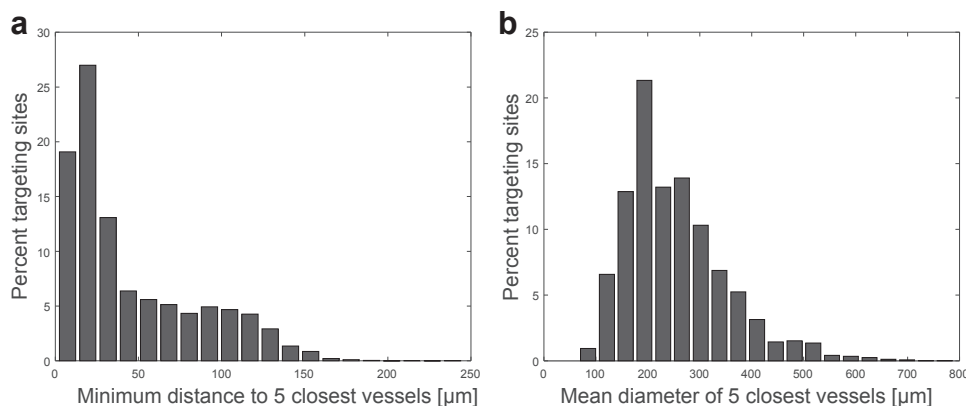


Fig. 8. Histograms of (a) the distance to the five closest resolvable vessels and (b) the mean diameter of these vessels in five animals indicate that most targeting occurred close to a vessel approximately 100–300 μm in diameter.

Clinical translation of the described technique would be aided by developments in both scanning system and contrast agent technology. Although there are no fundamental limitations that prevent the mechanically steered dual-frequency transducer from being used clinically for superharmonic imaging, a dual-frequency array transducer would allow for greater control over the pressure field, ensuring a more uniform point spread function over a larger field of view while maintaining the contrast-to-tissue ratio and spatial resolution of the presented images. Clinical use of ultrasound molecular imaging is presently limited by the lack of FDA-approved targeted microbubble contrast agents; however, clinically translatable contrast agents have recently been developed, such as BR55 from Bracco (Pochon *et al.* 2010; Pysz *et al.* 2010). Once targeted contrast agents are developed and approved, the described technique for superharmonic molecular imaging could enable ultrasound molecular imaging with higher resolution in all the applications currently under investigation including angiogenesis (Pochon *et al.* 2010); inflammation (Wang *et al.* 2013); and cancers of the breast (Bachawal *et al.* 2013), colon (Wang *et al.* 2015a) and prostate (Tardy *et al.* 2010). Combining a higher resolution molecular image with a registered angiography-like image provides additional information on microvascular morphology relative to the previously attained ultrasound molecular images for these applications, in which molecular images are displayed against a background B-mode image.

In addition, the ability to visualize both microvascular anatomy and molecular targeting may be of clinical use by providing currently unavailable information on the growth and extent of an individual tumor. For example, imaging $\alpha_v\beta_3$ expression has previously been accomplished using PET and SPECT in tumors of the breast and brain (Bach-Gansmo *et al.* 2006; Beer *et al.* 2008; Schnell *et al.* 2009), and also using near-infrared fluorescence imaging in colorectal tumors (Verbeek *et al.* 2014). The PET imaging studies utilized an ^{18}F -cyclic RGD tracer and then validated imaging results using post-resection immunohistochemistry, finding that while $\alpha_v\beta_3$ imaging is effective at locating tumors, its expression is “highly variable” (Beer *et al.* 2008; Schnell *et al.* 2009). While this heterogeneous expression of $\alpha_v\beta_3$ and the poor spatial resolution of PET make it difficult to use PET alone as either a screening test or a surgical planning tool, the high-resolution 3-D vascular and targeting information provided by acoustic angiography could make for an excellent tool for surgical planning after initial tumor localization. In this case, the expression of $\alpha_v\beta_3$ on both endothelial and tumor cells, as identified in prior PET studies (Beer *et al.* 2008), could prove helpful for ensuring resection with clean margins.

CONCLUSIONS

We have presented an ultrasound-based approach for both imaging vasculature and molecular markers with high spatial resolution. This technique uses a prototype dual-frequency transducer to acquire only superharmonic signals produced by microbubble contrast agents. Although the imaging process was observed to produce decreasing image brightness over time *in vitro* due to destruction of bound microbubbles, combinations of transmit pressure and frequency can be chosen that prolong microbubble persistence. Under such parameters, diagnostically useful superharmonic molecular images may be acquired. In this study, we have found high-resolution superharmonic molecular imaging in a rodent tumor model (fibrosarcoma) using microbubbles targeted to the $\alpha_v\beta_3$ integrin. A second scan of free microbubbles was acquired immediately after the molecular scan, and the two imaging volumes were fused to provide 3-D visualization of both microvascular anatomy and bio-distribution of the $\alpha_v\beta_3$ integrin. Mapping both biomarkers and microvascular anatomy with high resolution provides a new method for visualizing molecular markers with ultrasound while using ultrasound-based angiography-like mapping to provide anatomic information. The combined information provided by these scans may present new opportunities for analyzing relationships between microvascular anatomy and vascular targets for microbubbles such as integrins, selectins and other extracellular binding domains.

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